CIE AS PHYSICS (9702) • QUESTION PAPER

11.1 Atoms, Nuclei & Radiation

Syllabus Section: 11. Particle Physics **Paper Type:** Structured Questions (Paper 2) **Total:** 8 Questions (66 Marks)

Question 1 Easy [10 marks]

Part (a) [4 marks]

Ionising radiation often comes in three forms: alpha, beta and gamma.

Table 1.1 below shows information about the properties of each type of radiation.

Type	Nuclear notation	Relative charge
Alpha		+2
Beta	$^{0}_{-1}\beta$	
Gamma		

Table 1.1

Complete the missing information in Table 1.1.

Part (b) [2 marks]

Identify **one** similarity and **one** difference between alpha particles and beta particles**.**

Part (c) [1 mark]

The properties of ionising radiation that make it so useful is related to the study of the structure of atoms.

Evidence that atoms had an internal structure was confirmed by Ernest Rutherford. He was able to show that a thin sheet of gold foil scattered ionising radiation in ways that could not be explained by accepted models of atomic structure at the time.

State the type of ionising radiation that was scattered by gold foil in Rutherford's experiment.

Part (d) **[3 marks]**

Each of the observations from Rutherford's experiment had profound implications about how atoms were structured.

In the space provided below, match each observation from Rutherford's experiment to its corresponding conclusion about atomic structure.

Observation		Conclusion
A small proportion of radiation is slightly deflected by the gold foil	The atom is mostly empty space	
The majority of radiation is undeflected by the gold foil	The mass of a nucleus is concentrated in a very small volume	
A very small proportion of radiation is deflected back on itself	The nucleus carries a positive charge	

Question 2 Easy [8 marks]

Part (a) [2 marks]

The equation below represents the decay of tungsten-184.

Complete the equation by writing the correct number in each of the **two** boxes.

$$^{184}_{74}\mathrm{W} \rightarrow {}^{\square}_{\square}\mathrm{Po} + \text{alpha particle}$$

Part (b) [3 marks]

Beta decay does **not** cause the mass number of an atom to change.

Explain why not.

Part (c) [2 marks]

The equation representing the decay of iodine-128 to xenon is shown below.

Complete the equation by writing the correct number in each of the **two** boxes.

$$^{128}_{53}\mathrm{I} \rightarrow {}^{\square}_{\square}\mathrm{Xe} + \text{beta particle}$$

Part (d) [1 mark]

State another particle emitted in beta minus decay.

Question 3 Medium [3 marks]

Part (a) [2 marks]

A nucleus X decays by emitting a β+ particle to form a new nucleus, $^{23}_{11}Na$.

State the number of protons and the number of neutrons in nucleus X.

number of protons = ...

number of neutrons = ...

Part (b) [1 mark]

State **one** similarity between a β^+ particle and a β^- particle.

Question 4 Medium [11 marks]

Part (a) [2 marks]

The deflection of α-particles by a thin metal foil is investigated with the arrangement shown in Fig 1.1.

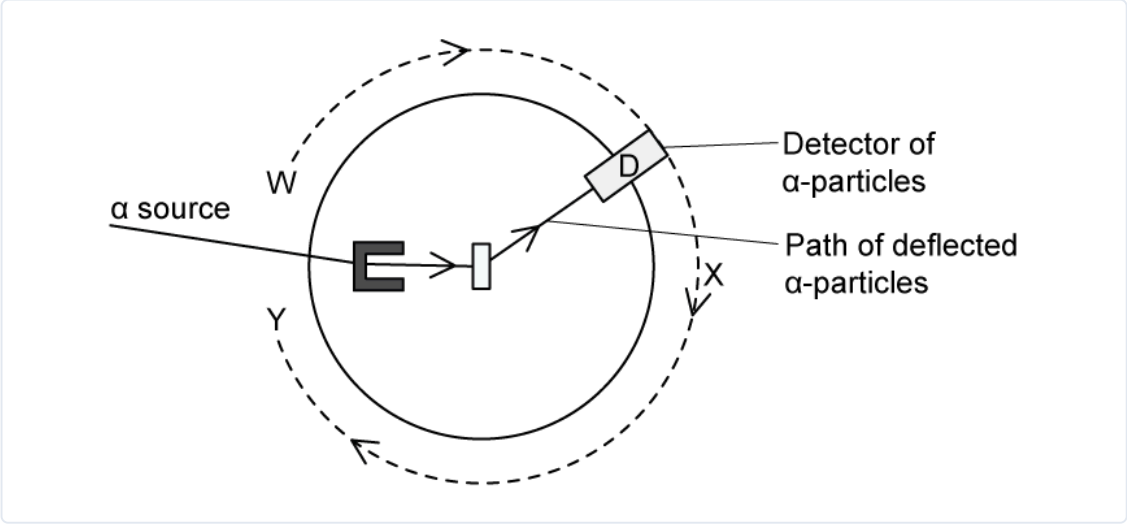


Fig 1.1

The detector of α-particles, D, is moved around the path labelled WXY.

State the environment the apparatus must be enclosed in and explain why this is required for the experiment.

Part (b) [4 marks]

Explain the readings detected by D and the conclusions that were made about the nature of atoms at points

(i) W,

[2]

(ii) X.

[2]

Part (c) [2 marks]

State what the readings detected by D at position Y conclude about the charge of the nucleus. Explain why.

Part (d) [3 marks]

A beam of α-particles produces a current of 3.5 pA. Calculate the number of α-particles per second passing a point in the beam.

Question 5 Medium [11 marks]

Part (a) [3 marks]

When a nucleus of plutonium-239 absorbs a neutron, the following reaction may take place.

$$^{239}_{94}\text{Pu} + {}^{a}_{b}\text{n} \rightarrow {}^{x}_{54}\text{Xe} + {}^{103}_{y}\text{Zr} + 3\,{}^{a}_{b}\text{n}$$

State the values of *a*, *b*, *x* and *y*.

Part (b) [2 marks]

State two quantities, other than the proton and neutron number, that are conserved in the fission reaction in part (a).

Part (c) [3 marks]

A Plutonium-239 nucleus has a mean density of 1.3×10^{17} kg m^{-3}.

Show that the radius of a Plutonium-239 nucleus is 9.0×10^{-15} m.

Part (d) [3 marks]

Plutonium-239 is an α-emitter whilst Uranium-237 is a β-emitter.

Aside from their structure, distinguish between an α-particle and a β-particle.

Question 6 Medium [3 marks]

Part (a) [3 marks]

In the alpha-particle scattering experiment, a beam of alpha particles was directed at a thin gold foil.

State the conclusion that can be drawn about the structure of the atom from

(i) the observation that most alpha-particles pass through the foil

[1]

(ii) the observation that very few alpha-particles were deflected through angles greater than 90°.

[2]

Question 7 Hard [10 marks]

Part (a) [3 marks]

The study of beta decay provided early hints to the existence of a yet undiscovered particle.

Scientists were interested in the energy distribution of the beta particles emitted when a radioactive source decayed. The source has an activity of 120 Bq.

The energy released by each decaying nucleus was labelled as X and the total energy released each second by the source was measured to be 7.5×10^{-11} J.

Sketch a graph to show the energy distribution of the beta particles and indicate X on the diagram.

Part (b) [2 marks]

Determine the value of X to an appropriate degree of accuracy.

Part (c) [2 marks]

Explain why no beta particles had an energy equal to X.

Part (d) [3 marks]

Suggest the standard form of the two possible equations that represent this decay and define all the terms used.

Question 8 Hard [10 marks]

Part (a) [2 marks]

A radioactive nucleus $^{229}_{85}\mathrm{X}$ undergoes a beta−minus decay followed by an alpha decay to form a daughter nucleus $^{A}_{Z}\mathrm{Y}$.

Write a decay equation for this interaction and hence determine the values of A and Z.

Part (b) [3 marks]

Thorium, $^{232}_{90}\mathrm{Th}$ decays to an isotope of Radium (Ra) through a series of transformations. The particles emitted in successive transformations are:

$$\alpha \quad \beta \quad \beta \quad \gamma \quad \alpha$$

Determine the resulting nuclide after these successive transformations.

Part (c) [3 marks]

Through a combination of successive alpha and beta decays, the isotope of any original nucleus can be formed.

Explain the simplest sequence of alpha and beta decays required to do this.

Part (d) **[2 marks]**

A nucleus of Bohrium $^{X}_{Y}\mathrm{Bh}$ decays to Mendelevium $^{255}_{101}\mathrm{Md}$ by a sequence of three alpha particle emissions.

Determine the number of neutrons in a nucleus of $^{X}_{Y}\mathrm{Bh}$.